Quantization of Diffusion Models

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Abstract

Diffusion models provide high-quality image generation but require substantial memory and computational resources. This project studies post-training weight quantization of Stable Diffusion 1.5, with a focus on activation-aware vector quantization. We investigate residual vector quantization as a computationally simpler alternative to additive vector quantization with joint beam-search code assignment. The considered improvements include stage-wise allocation of codebook sizes, orthogonal Hadamard transforms, and post-quantization refinement of discrete codes. Preliminary experiments on the middle block of the U-Net show that the resulting method approaches the quality of the strong VQDM baseline while avoiding joint beam-search assignment and requiring no fine-tuning of the diffusion model. Further work will extend the evaluation to the full U-Net, isolate the contribution of each component, and use larger-scale generation-quality metrics.

Keywords: TODO

1 Introduction

TODO

Contributions. Our contributions can be summarized as follows:

- We present...
- We demonstrate the validity of our theoretical results through empirical studies...
- We highlight the implications of our findings for...

Outline. The rest of the paper is organized as follows...

2 Related Work

Topic #1. TODO

Topic #2. TODO

39th Conference on Neural Information Processing Systems (NeurIPS 2025).

3 Preliminaries

3.1 General notation

In this section, we introduce the general notation used in the rest of the paper and the basic assumptions.

3.2 Assumptions

TODO

4 Method

5 Experiments

To verify the theoretical estimates obtained, we conducted a detailed empirical study...

6 Discussion

TODO

7 Conclusion

TODO

A Appendix / supplemental material

A.1 Additional experiments / Proofs of Theorems

TODO